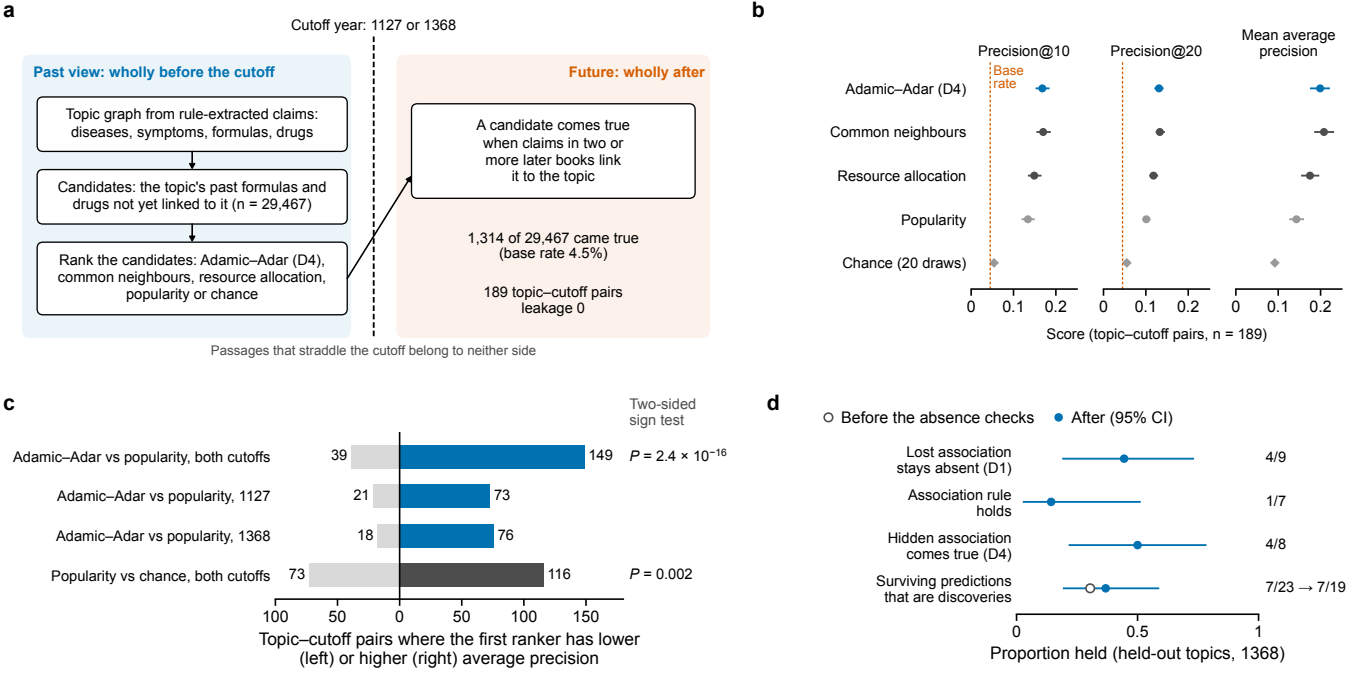
a
Cutoff year: 1127 or 1368
Past view: wholly before the cutoff
Topic graph from rule-extracted claims: diseases, symptoms, formulas, drugs
Candidates: the topic's past formulas and drugs not yet linked to it (n = 29,467)
Rank the candidates: Adamic–Adar (D4), common neighbours, resource allocation, popularity or chance
Future: wholly after
A candidate comes true when claims in two or more later books link it to the topic
1,314 of 29,467 came true (base rate 4.5%)
189 topic–cutoff pairs
leakage 0
Passages that straddle the cutoff belong to neither side
b
Precision@10
Precision@20
Mean average precision
Base rate
Adamic–Adar (D4)
Common neighbours
Resource allocation
Popularity
Chance (20 draws)
0
0.1
0.2
Score (topic–cutoff pairs, n = 189)
c
Two-sided sign test
Adamic–Adar vs popularity, both cutoffs
39
149
P = 2.4 × 10−16
Adamic–Adar vs popularity, 1127
21
73
Adamic–Adar vs popularity, 1368
18
76
Popularity vs chance, both cutoffs
73
116
P = 0.002
100
50
0
50
100
150
Topic–cutoff pairs where the first ranker has lower (left) or higher (right) average precision
d
Before the absence checks
After (95% CI)
Lost association stays absent (D1)
4/9
Association rule holds
1/7
Hidden association comes true (D4)
4/8
Surviving predictions that are discoveries
7/23 → 7/19
0
0.5
1
Proportion held (held-out topics, 1368)